# TANEL TREUBERG


Software Engineer

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## PROFESSIONAL SUMMARY

Software Engineer with 3 years of experience in backend development, infrastructure automation, and applied machine learning. Proven track record of delivering measurable performance improvements in energy-sector web applications and building production computer vision pipelines for autonomous marine vessels. Proficient in Python, Docker, and CI/CD, with a strong foundation in embedded ML deployment.

## WORK EXPERIENCE

**Elering** Jul 2022 – Jun 2024

*Software Engineer*

- Refactored an internal web application (backend and frontend) that pulls data from the SCADA ISR database, reducing data export time by 10x and cutting average UI load time from ~8s to under 2s.
- Built and maintained Python automation scripts executed via GitLab CI/CD, eliminating ~15 hours/month of manual data processing across operations teams.
- Developed and tested features for the internal Balancing Market System (BMS), reducing message delivery latency and processing time by 30% for Estonia's national energy grid operations.
- Containerised monolithic applications using Docker, enabling consistent deployments across development and test environments.
- Managed Jenkins-based process automations with monitoring via Grafana and ELK stack, reducing incident detection time for critical pipelines.

**Baltic WorkBoats** Nov 2021 – Jul 2022

*Software Engineer — Computer Vision & Embedded ML*

- Designed and deployed a real-time computer vision system for an autonomous Navy patrol vessel (Navy 18 WP), enabling automated maritime object detection for unmanned maneuvering trials.
  - Selected YOLOv5 (Ultralytics) for optimal speed-accuracy trade-off; containerised the full pipeline with Docker on Nvidia NGC for reproducible deployment on Jetson AGX Xavier (32 GB).
- Optimised inference pipeline using TensorRT FP16 quantisation, ONNX conversion, and NVDEC hardware video decoding, achieving an average 3.5x speed increase (best case 4.3x).
  - Negligible accuracy loss ($1.7 \times 10^{-3}$ mAP), 26.8% lower energy consumption, and 3.3°C temperature reduction, enabling sustained autonomous operation at sea.
- Built multi-camera support (3 simultaneous inputs via GStreamer + OpenCV), reaching 1.4x throughput over single-input at the same resolution, with architecture-aware code paths for x86_64 dev and aarch64 production.

**SP Engineers OÜ** Jul 2021 – Oct 2021

*Electrical Engineering Intern*

- Developed firmware for ESP32 microcontrollers powering sensor-aided monitoring prototypes, gaining hands-on experience with embedded C++ and hardware-software integration.

## EDUCATION

**Tallinn University of Technology** Graduated Jun 2023

*BSc in Engineering — Mechatronics (Electroenergetics & Mechatronics)*

- Thesis: "Improving Situational Awareness of Autonomous Vessels Using Computer Vision" — developed and benchmarked 10 YOLOv5 model variants on Jetson AGX Xavier for real-time maritime object detection.
- Organised TalTech Energy Conference 2020; former member of Faculty of Engineering Student Council (INSÜK).

## SKILLS & INTERESTS

**Languages:** Python, C++, JavaScript, SQL, Bash, Swift
**ML / AI:** PyTorch, TensorFlow, TensorRT, NumPy, Pandas, Nvidia Jetson
**Infrastructure & DevOps:** Docker, Jenkins, GitLab CI/CD, AWS S3, Grafana, ELK, Kubernetes (basic)
**Frameworks & Tools:** Flask, REST API, Selenium, Scrapy, PostgreSQL, Postman, Git, Jira, Confluence
**Platforms:** GNU/Linux, macOS
**Interests:** Social dancing (tango argentino, Brazilian zouk), guitar and piano, physics